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EXERCISE 2.1

1. The graphs of $y = p(x)$ are given in Fig. 2.10 below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$, in each case.

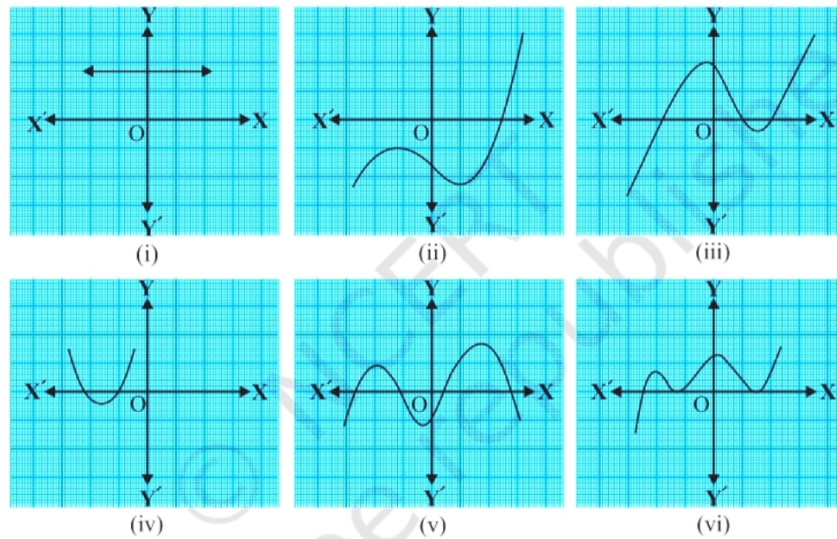


Fig. 2.10

Fig. 2.10

Example 4: Find a quadratic polynomial, the sum and product of whose zeroes are -3 and 2 , respectively.

Solution: Let the quadratic polynomial be $ax^2 + bx + c$ and its zeroes be α and β .

We have $\alpha + \beta = -3 = -\frac{b}{a}$ and $\alpha\beta = 2 = \frac{c}{a}$.

If $a = 1$, then $b = 3$ and $c = 2$.

So, one quadratic polynomial which fits the given conditions is $x^2 + 3x + 2$.

You can check that any other quadratic polynomial which fits these conditions will be of the form $k(x^2 + 3x + 2)$, where k is a real number.

Let us now look at cubic polynomials. Do you think a similar relation holds between the zeroes of a cubic polynomial and its coefficients?

Let us consider $p(x) = 2x^3 - 5x^2 - 14x + 8$.

You can check that $p(x) = 0$ for $x = 4, -2$ and $\frac{1}{2}$. Since $p(x)$ can have at most three zeroes, these are the zeroes of the given polynomial.

Now, the sum of the zeroes is $4 + (-2) + \frac{1}{2} = \frac{5}{2} = -\frac{\text{coefficient of } x^2}{\text{coefficient of } x^3}$.

The product of the zeroes is $4 \times (-2) \times \frac{1}{2} = -4 = -\frac{\text{constant term}}{\text{coefficient of } x^3}$.

The sum of the products of the zeroes taken two at a time is

$$(4 \times -2) + (-2 \times \frac{1}{2}) + (\frac{1}{2} \times 4) = -7 = \frac{\text{coefficient of } x}{\text{coefficient of } x^3}.$$

In general, if α, β and γ are the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}.$$

Example 5*: Verify that 3, -1 and $-\frac{1}{3}$ are the zeroes of the cubic polynomial $p(x) = 3x^3 - 5x^2 - 11x - 3$, and then verify the relationship between the zeroes and the coefficients.

Solution: Comparing the given polynomial with $ax^3 + bx^2 + cx + d$, we get $a = 3$, $b = -5$, $c = -11$ and $d = -3$.

Further,

$$p(3) = 3(3)^3 - 5(3)^2 - 11(3) - 3 = 81 - 45 - 33 - 3 = 0$$

$$p(-1) = 3(-1)^3 - 5(-1)^2 - 11(-1) - 3 = -3 - 5 + 11 - 3 = 0$$

$$p\left(-\frac{1}{3}\right) = 3\left(-\frac{1}{3}\right)^3 - 5\left(-\frac{1}{3}\right)^2 - 11\left(-\frac{1}{3}\right) - 3 = -\frac{1}{9} - \frac{5}{9} + \frac{11}{3} - 3 = 0$$

Therefore, 3, -1 and $-\frac{1}{3}$ are the zeroes of the given polynomial.

So, we take $\alpha = 3$, $\beta = -1$ and $\gamma = -\frac{1}{3}$.

Now,

$$\alpha + \beta + \gamma = 3 + (-1) + \left(-\frac{1}{3}\right) = \frac{5}{3} = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = (3)(-1) + (-1)\left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right)(3) = -3 + \frac{1}{3} - 1 = -\frac{11}{3} = \frac{c}{a}$$

$$\alpha\beta\gamma = (3)(-1)\left(-\frac{1}{3}\right) = 1 = -\frac{d}{a}$$

Hence, the relationship between the zeroes and the coefficients of the given cubic polynomial is verified.

Example 6: Divide $2x^2 + 3x + 1$ by $x + 2$.

Solution: We stop the division process when either the remainder is zero or its degree is less than the degree of the divisor.

Here, the quotient is $2x - 1$ and the remainder is 3.

Also, $(2x - 1)(x + 2) + 3 = 2x^2 + 3x + 1$.

That is, $2x^2 + 3x + 1 = (x + 2)(2x - 1) + 3$.

Therefore, Dividend = Divisor * Quotient + Remainder.

Let us now extend this process to divide a polynomial by a quadratic polynomial.